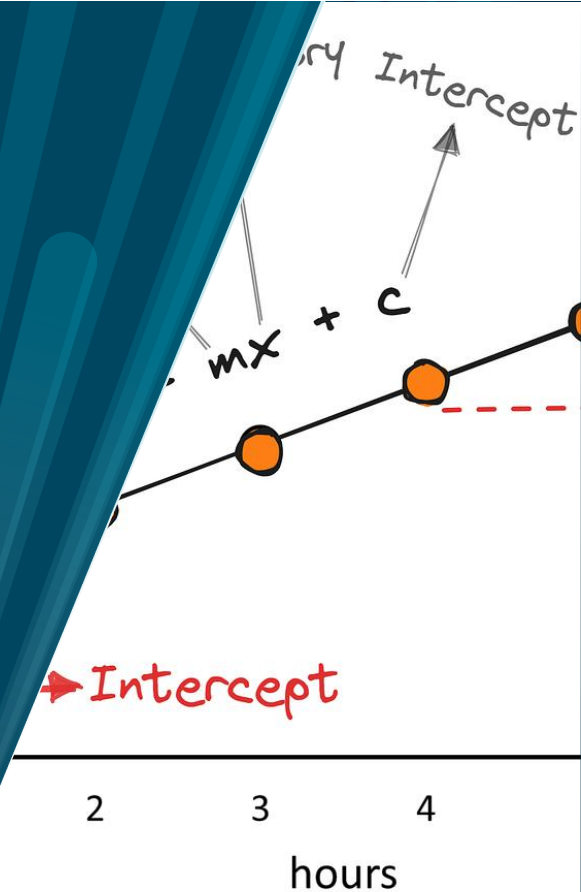


Chapter 2

The Simple Linear Regression Model

Sok-3081

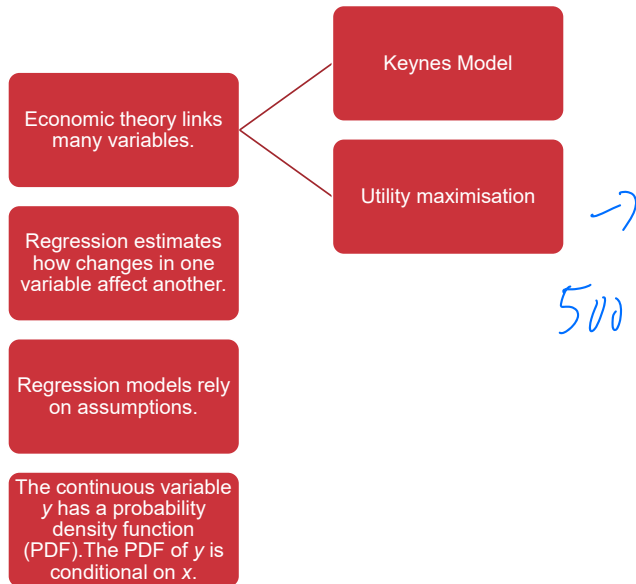
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9. September 2026



Learning path

- Economic question
 - Regression model
 - OLS line
 - Coefficients and residuals
 - Estimator and estimate
 - Repeated samples
 - Precision and standard errors
- 

An Economic Model



Think of typical food expenditure

- Does increase or decrease with income?
- How much do you think is fixed?
- How much depend on income?

0.15

If income in the region increases by \$100 dollars much more sales does local retailers get?

An economic model

How does weekly food expenditure vary with household income?



Economic theory suggests a positive relationship.



A hypothetical rule: food expenditure = $80 + 0.12 \times \text{income}$.



Here, both expenditure and income are measured in dollars.

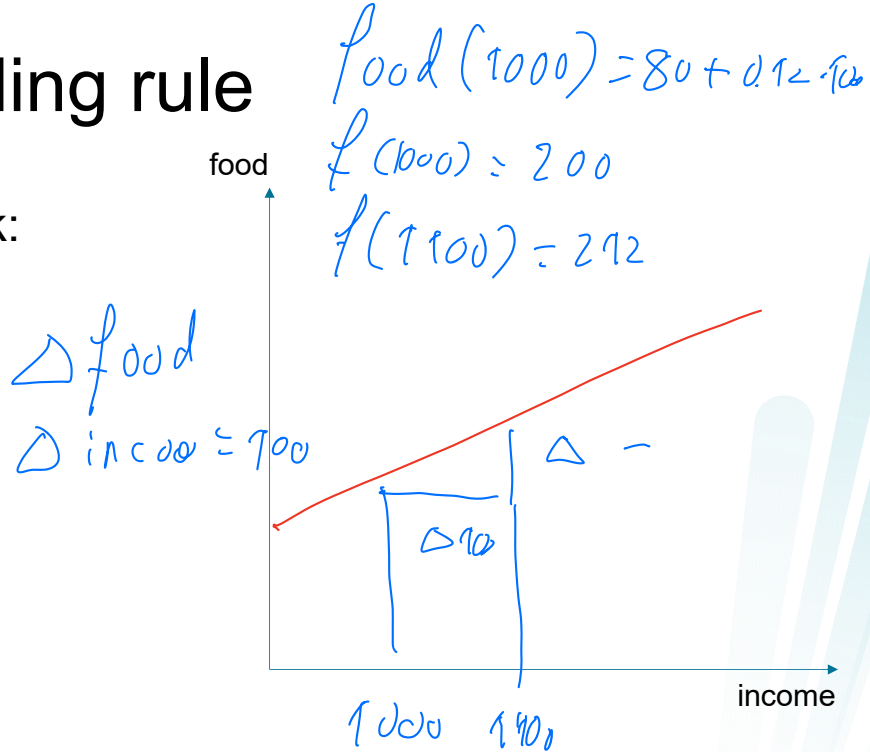
A household spending rule

A household spends per week:

- 80 dollars
- plus 12 cents from every dollar of income

Income rises from 1,000 to 1,100 dollars

How much does food expenditure change?



A simple model

- Let us add our simple model

$$food = 80 + 0.12 \cdot income$$

- Okey fit
 - But not very scientific



The sample and its units

- The sample contains 40 households.
 - 3 persons household
- y = weekly food expenditure, measured in dollars.
- x = weekly household income, measured in dollars.
- From the dataset food

The simple regression model

$$y_i = \underbrace{\beta_1 + \beta_2 \cdot x_i}_{\text{Economic model}} + \underbrace{e_i}_{\text{Everything else}}$$

$$E(y | x) = \mu_{y|x}$$

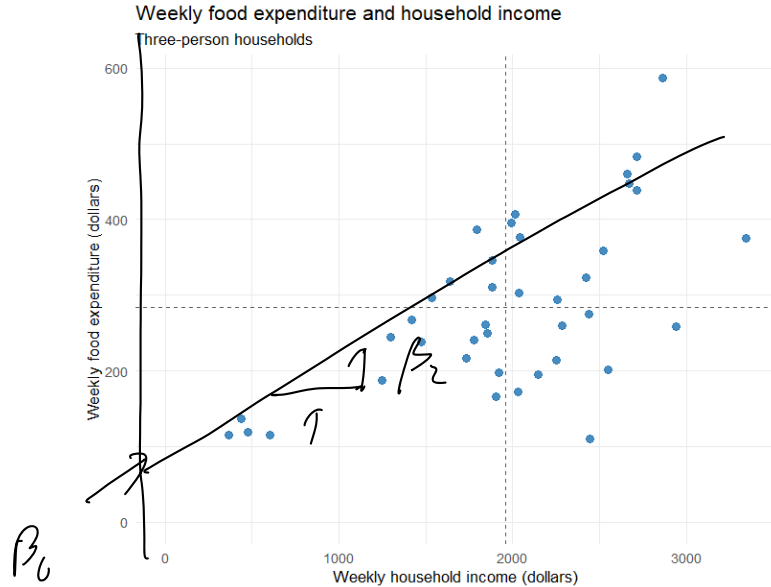


Figure 2.1

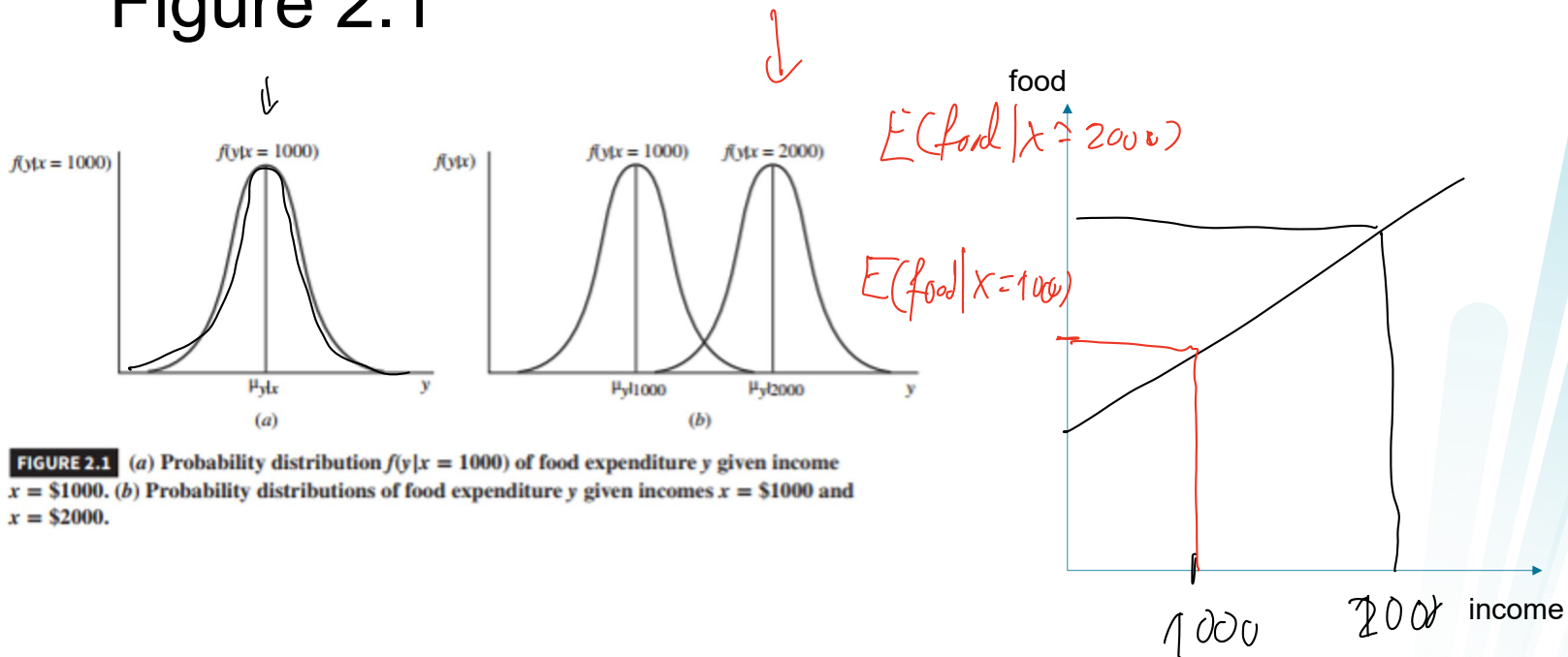


FIGURE 2.1 (a) Probability distribution $f(y|x = 1000)$ of food expenditure y given income $x = \$1000$. (b) Probability distributions of food expenditure y given incomes $x = \$1000$ and $x = \$2000$.

The econometric model

$$y_i = \beta_1 + \beta_2 \cdot x_i + e_i$$

Population parameters:

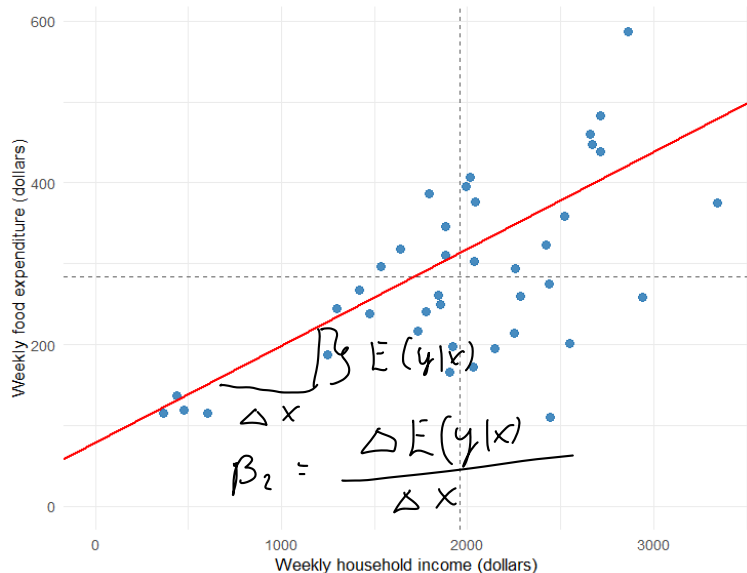
- β_1 is the population intercept.
- β_2 is the population slope.

Error term:

- e_i includes other influences on y_i that the model does not explicitly include.
- We observe: x_i and y_i .
- We do not observe β_1 , β_2 , or e_i .

$$y_i = E(y_i | x)$$

Weekly food expenditure and household income
Three-person households



The population regression function

$$E(y_i|x_i) = \beta_1 + \beta_2 \cdot x_i \quad \text{when} \quad E(e_i|x_i) = 0$$

- The regression function describes average y at a given x .
- β_2 is the change in that conditional mean for a one-unit increase in x .
- Individual outcomes can lie above or below the conditional mean.

The simple regression assumptions

- SR 1: $y_i = \beta_1 + \beta_2 \cdot x_i + e_i$ 
- SR 2: $E(e_i|x) = 0$ 
- SR 3: $var(e_i|x) = \sigma^2$ for every observation. 
- SR 4: $cov(e_i, e_j|x) = 0$ for $i \neq j$
- SR 5: The sample contains at least two distinct x values. 
- SR 6: Error Normality $e_i|x \sim N(0, \sigma^2)$, optional 

SR1: The econometric model

$$y_i = \beta_1 + \beta_2 \cdot x_i + e_i$$

What does this tell us?

- y is determined by a systematic part and a random part
- x explains some of the variation in y
- everything else affecting y is collected in e_i

SR2: Strict exogeneity

$$E(e_i|x) = 0$$

What does this tell us?

- omitted factors are not systematically related to x
- at each value of x , positive and negative errors average to zero
- this is the key assumption for interpreting β_2 as the effect of x on y

SR3: Homoskedasticity

$$E(y|x) = \mu_{y|x}$$

$$\text{var}(e_i|x) = \sigma^2$$

What does this tell us?

- the amount of unexplained variation is the same for all values of x
- observations are equally spread around the regression line across x

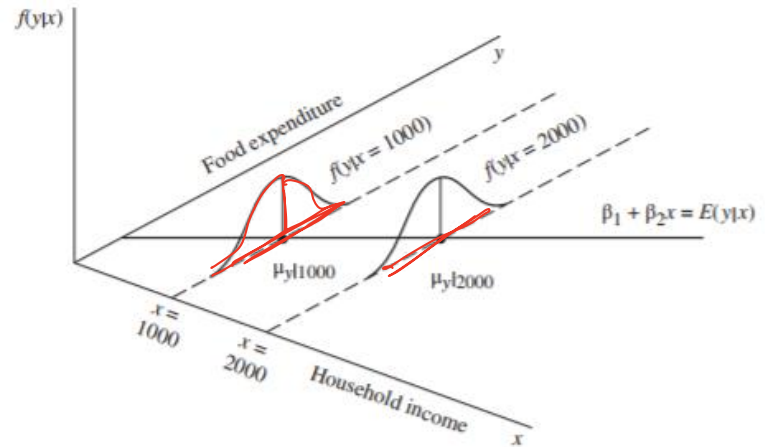


FIGURE 2.5 The conditional probability density functions for y , food expenditure, at two levels of income.

SR4: Uncorrelated errors

$$\text{cov}(e_i, e_j | x) = 0 \text{ for } i \neq j$$

What does this tell us?

- the unexplained part of one observation is not related to the unexplained part of another
- an error for observation i gives us no information about the error for observation j

SR5: Variation in x

What does this tell us?

- we need different values of x in the sample
- without variation in x , we cannot estimate how y changes when x changes



SR6: Normality



$$e_i|x \sim N(0, \sigma^2)$$

What does this tell us?

- the unexplained part of one observation is not related to the unexplained part of another
- an error for observation i gives us no information about the error for observation j

Theoretical $\rightarrow$ Estimated $\mathcal{U}_{\mu, \sigma} \Rightarrow \overline{X}_{\mu, \sigma} \quad b_1$

- When we start estimating parameters change

$$y_i = \beta_1 + \beta_2 \cdot x_i + e_i$$



$$\hat{y}_i = b_1 + b_2 \cdot x_i$$

$$\hat{e}_i = y_i - \hat{y}_i$$

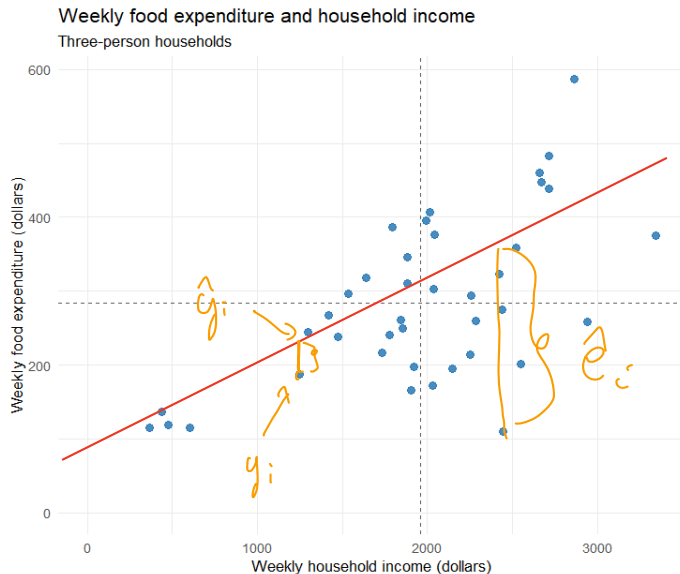
Theoretical / Population	Estimated / Observed	Meaning
y_i	$\hat{y}_i$	Observed outcome / Fitted value
β_1, β_2	b_1, b_2	Population parameters / OLS estimates
e_i	$\hat{e}_i$	Random error / Residual

Ordinary least squares

$$\hat{e}_i = y_i - \hat{y}_i$$

$$\hat{y}_i = b_1 + b_2 \cdot x_i$$

- The sample estimates b_1 and b_2 determine the fitted line.
- The fitted value $\hat{y}_i$ estimates the average outcome at x_i .
- OLS selects the intercept and slope that minimize squared residuals.



Fitted values and residuals

$$\hat{e}_1^2 + \hat{e}_2^2 + \hat{e}_3^2 + \dots + \hat{e}_n^2$$

$$\hat{e}_i = y_i - \hat{y}_i$$

- A residual is the observed outcome minus the fitted value.
- A positive residual means the observation lies above the fitted line.
- The residual $\hat{e}_i$ estimates an unobserved error e_i .



Fitted values and residuals

$$SSE = \sum_i (y_i - b_1 + b_2 \cdot x_i)^2$$

- A residual is the observed outcome minus the fitted value.
- A positive residual means the observation lies above the fitted line.
- The residual $\hat{e}_i$ estimates an unobserved error e_i .

$$\hat{e}_i = y_i - \hat{y}_i$$

$$\hat{y}_i = b_1 + b_2 \cdot x_i$$

$$\text{Min} \sum (\hat{e}_i)^2$$

$$\text{Min}_{b_1, b_2} \sum (y_i - b_1 - b_2 \cdot x_i)^2$$

The OLS formulas

$$b_1 = \bar{y} - b_2 \cdot \bar{x}$$

$$b_2 = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2}$$

- If all x values are equal, the slope denominator is zero.
- The slope depends on how x and y vary together, relative to variation in x.

$$b_1 = \bar{y}$$

$$\sum_i (x_i - \bar{x})^2 = 0$$

$$\sum (y_i - \bar{y}) = 0 \quad b_2 = 0$$

$$\frac{\text{Cov}(x, y)}{\text{Var}(x)}$$


$$\text{Cov}(x, y) = 0 \Rightarrow b_2 = 0$$

$$b_1 = \bar{y} - \underbrace{b_2 \cdot \bar{x}}_0$$

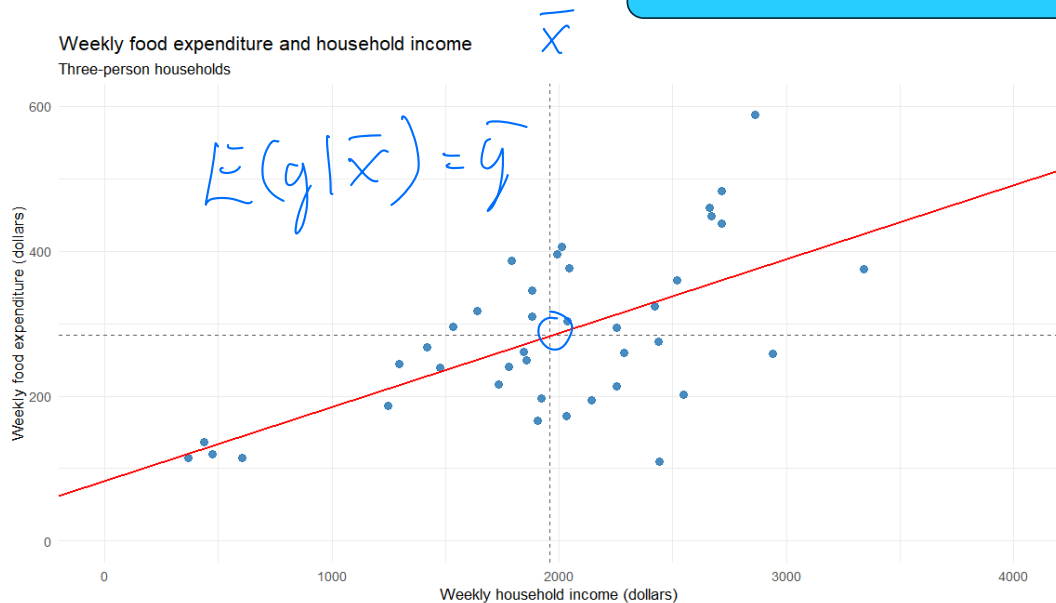
OLS vs our simple model

- Estimate the OLS model using formulas
- Run `lm()` function
- Find $\sum_i \hat{e}_i^2$ and mean $\bar{\hat{e}_i}$ for:
 - Our simple model
 - OLS
 - `Lm()`

```
R  
lm(food_exp ~ income, data = food)
```


Plotting OLS

$$\hat{y}_i = 83.416 + 10.209 \cdot x_i$$



Properties of the fitted line

- The fitted line passes through the sample means: $b_1 = \bar{y} - b_2 \cdot \bar{x}$
- The residuals sum to zero: $\sum_i \hat{e}_i = 0$
- The regressor and residuals have zero sample covariance.
- These algebraic properties do not establish population exogeneity.

Prediction at a given income

$$\Delta x_i = 100$$

$$E(y, \Delta x = 100) \\ \beta_2 \cdot \Delta x$$

$$0.10209 \cdot 100 = 10.2$$

$$\hat{y}_i = 83.416 + 10.209 \cdot x_i$$

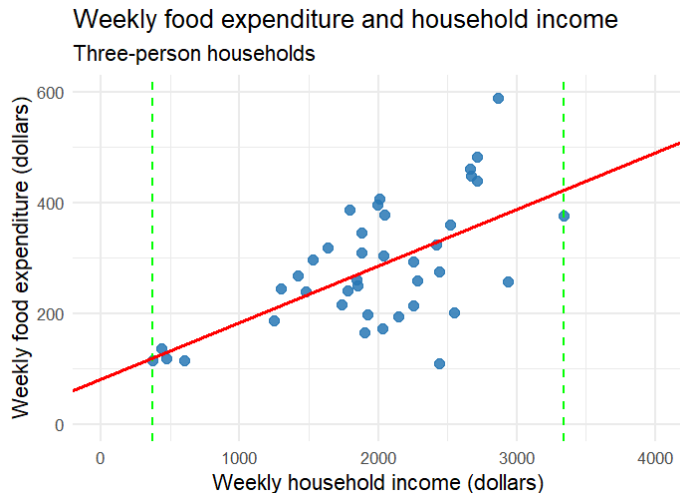
- $x = 3000$ corresponds to a weekly income of \$3,000
- An individual household can spend more or less than the fitted value.



$$\hat{y}_i = 83.416 + 10.209 \cdot 3000 = 389.7$$

Avoid extrapolation!

- Predictions are most credible **within the observed range of x** .
- Outside this range, we assume the same linear relationship continues.
- That assumption may be unrealistic.



Elasticise

- Using income in dollars changes the estimated slope to 0.1021.
- The fitted values remain identical when the unit conversion is correct.
- For a linear model, elasticity at $(x, \hat{y})$ is $b_2 \times x / \hat{y}$.
- At the sample means, the food-expenditure elasticity is about 0.706.

Elasticity and income

$$\hat{\varepsilon}(x) = b_2 \frac{x}{\hat{y}}$$

- Food expenditure increases with income, but less than proportionally.
- Elasticity increases with income, but at a decreasing rate.
- In our sample, elasticity remains below 1 $\rightarrow$ food is a necessity.
- At the same time, food expenditure becomes a smaller share of income as income rises.

